Amagi-MEM v1.0

Self-Protecting, Context-Aware Memory Architecture for High-Assurance Systems

Author: Alexey Mikhailovich Burlai

ORCID: 0009-0001-4679-5967

Affiliation: Independent Researcher, Sochi, Russia

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Contact: lexintrud@gmail.com

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0. Abstract

Amagi-MEM v1.0 is not a general-purpose memory architecture.

It is a high-assurance, domain-specific substrate for systems where data integrity, contextual authenticity, and system-level safety outweigh raw performance.

Built on the **TTP principle** (Trigger → Programmable Delay → Localized Activation), it embeds the six Core Principles of the Amagi Framework v1.0 into a **physically isolated, self-securing memory fabric**.

This specification defines a **feasible architecture** for deployment in:

- **Medical implants** (e.g., AI-driven pacemakers, neurostimulators),
- **Aerospace and defense systems** (avionics, satellites),
- **Financial infrastructure** (next-gen HSMs, central banking),
- **Critical industrial control** (nuclear, power grid),
- Autonomous AI agents with real-world agency —

  domains where failure is not an option.

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1. Scope and Intent

Amagi-MEM is not intended to replace DDR5, HBM, or LPDDR in consumer computing.

Its design explicitly trades latency for verifiable trust — a trade-off unacceptable for gaming or web browsing, but mandatory for life-critical and mission-critical systems.

The architecture assumes a **green-field hardware-software co-design environment**, where CPU, OS, and memory controller are developed in concert.

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2. Core Principles → Hardware Realization

Amagi Framework Principle	Amagi-MEM Realization
Immutability	MAG fused in OTP; audit logs append-only and Dilithium-signed
Constrained Communication	All data flows strictly defined by immutable MAG matrix
Contextual Oversight	MPE validates Source ID = CoreID ⊕ µCodeHash ⊕ TPM_PCR[17]
Controlled Evolution	No runtime updates; OTP-only microcode
Automatic Safety Preservation	ATL + SHMF trigger Safe State and self-isolation
Complete Audit Integrity	Every access cryptographically signed and immutable

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3. Architectural Layers (Physically Isolated)

3.1. Memory Policy Enforcer (MPE)

Hardware Root of Trust — Decision Engine

- Validates **Source ID** internally (never over bus).
- Queries **MAG** for policy.
- Issues **TTP command** to ATL.
- Logs all events to **immutable, signed audit buffer**.
- **Feasibility**: Standard in secure ASICs (e.g., medical SoCs).

3.2. Memory Access Graph (MAG)

Immutable Policy Matrix — 128×128 OTP

- One-time programmable at factory or first secure boot.
- 4-bit policy: `Deny` / `Immediate` / `Delayed` / `High-Delay` / `External Trigger`.
- Lookup in <5 ns via combinational logic.

- **Feasibility**: Used in secure enclaves (Apple, HSMs).

3.3. Access Trigger Layer (ATL)

TTP Engine — Delay as a Security Primitive

- Implements delay via **digital delay lines (DDL)** and **tunable RC buffers**.
- Range: **0.5 ns to 1 µs**, physically enforced.
- Output directly controls **power-gating transistors**.
- **Feasibility**: Acceptable in latency-tolerant domains (implants, avionics).

3.4. Self-Healing Memory Fabric (SHMF)

Active, Monitored Substrate

- **Row-level encryption**: AES-256 + Kyber (PQC hybrid).
- **Self-monitoring**: Voltage, temperature, error rates.
- **Self-isolation**: Powers down rows, migrates data to spares.
- **Feasibility**: Emerging in PIM and custom ASICs.

3.5. Amagi-Q Cryptographic Engine

Integrated PQC Co-Processor

- On-the-fly encryption/decryption.
- Dilithium signing of audit logs.
- Keys bound to hardware root of trust.
- **Feasibility**: PQC accelerators in FPGA/ASIC prototypes.

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4. End-to-End Access Flow

- Memory access request issued.
- MPE computes Source ID internally.
- MPE queries MAG[source][target].
- MAG returns 4-bit policy code.
- MPE sends TTP command to ATL.
- ATL enforces programmable delay via DDL/RC.
- ATL activates power gating for target row.
- SHMF decrypts data internally via Amagi-Q.
- Data released to system.
- MPE appends signed audit record (Dilithium).

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5. Feasibility Assessment

Component	Feasibility	Timeline	Domain
MPE + MAG	High	Now	Secure ASICs, HSMs
Amagi-Q (PQC)	Medium-High	1–3 years	Government, finance
SHMF	Medium	3–5 years	Medical, aerospace
ATL	Medium (niche)	Now	Implants, industrial
Full Integration	**High (specialized)**	**5–10 years**	Critical infrastructure

Bottleneck: Not technical feasibility, but **systemic adoption**:

- Co-design (CPU–memory–OS),
- Certification (ISO 26262, DO-178C, IEC 62304),
- Secure supply chain and ecosystem readiness.

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6. Derivative Architecture Clause (S-APL)

A **Derivative Architecture** is any system — regardless of name, technology, or stack — that implements **all six Core Principles of the Amagi Framework v1.0**, **as realized through the six hardware components**:

1. **Contextual Oversight** → MPE
2. **Constrained Communication** → MAG
3. **Controlled Evolution** → OTP-only MPE firmware
4. **Automatic Safety Preservation** → ATL + SHMF
5. **Immutability** → OTP MAG + signed logs
6. **Complete Audit Integrity** → Amagi-Q + append-only audit

Structural Pattern Protection: Reproducing the full sequence

Context → Policy → Delay → Activation → Self-Healing → Audit

is deemed Principle Extraction and requires an S-APL license.

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7. Conclusion

Amagi-MEM v1.0 does not seek to optimize the past.

It is designed for a future where memory must be a sovereign entity of trust.

It is not for your laptop.

It is for the AI that controls your pacemaker,

the HSM that secures a central bank,

the satellite that guides your aircraft,

and the robot that operates in a nuclear facility.

This is **Ethical Computing, architected for consequence-aware systems**.

Amagi-MEM v1.0 — Memory with Immunity, for Systems that Cannot Fail.

Appendix A: Formal Definitions

- **Target Domains**: Medical, aerospace, defense, finance, critical infrastructure, autonomous AI.
- **Not Target Domains**: Consumer electronics, general-purpose servers.
- **Co-Design Assumption**: CPU, OS, memory developed together.
- **Performance Model**: Bounded, predictable latency; throughput secondary.

Appendix B: License and Legal Provisions

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- Governed by **laws of England and Wales**.
- Contact: lexintrud@gmail.com

Note: The public S-APL template is for informational purposes only and does not grant rights.

Appendix C: Reference Diagrams

Two SVG files are included in this Zenodo record:

1. Amagi_MEM_Reference_Architecture.svg

   — Layered view: MPE → (ATL + MAG) → SHMF → Amagi-Q.

2. Amagi_MEM_TTP_Process.svg

   — TTP flow: Trigger → Programmable Delay (DDL/RC) → Localized Activation (Power Gating).

Amagi-MEM v1.0 — A Memory Architecture for the Age of Consequence.
